

Full Derivation of the Weltzeituhr Error Key Decryption Method for Kryptos K4

Ryan Bonifacino
Visiting Fellow
Massachusetts Institute of Technology
Sloan School of Management
ryan.bonifacino@sloan.mit.edu

November 18, 2025

Herbert James Sanborn Jr
Piney Point, Maryland

Dear Mr. Sanborn,

Please find the detailed derivation below:

1. Introduction

This document provides the complete mechanical derivation for the proposed decryption of Kryptos K4 using the **Weltzeituhr Error Key** - a running-key Vigenère system derived solely from historically documented changes to the Berlin World Time Clock.

No plaintext is assumed or guessed. All steps are deterministic and reproducible using:

- the 1969 Weltzeituhr city list,
- the 1997 repair/renaming records,
- the K4 ciphertext,
- and the OBKR header.

2. Cipher System

2.1 Structure

The cipher functions as a **running-key Vigenère variant**:

$$\text{Plain} = (\text{Cipher} - \text{Key}) \bmod 26$$

with letters mapped A=0, ..., Z=25.

2.2 Start Point

The header **OBKR** is interpreted as four offsets. For Row 3, the third letter **K** (11th letter) indicates:

- **Hour 11**
- **ENE sector** on the Weltzeituhr
- covering Soviet/Eastern Bloc cities

Thus the key begins at the **Russia block** of the clock.

3. Key Generation Rule (“Weltzeituhr Error Key”)

The running key is defined by a single rule:

Extract the initial letter of every city on the 1969 Weltzeituhr that was renamed, removed, or time-zone-shifted by 1997, in clockwise order starting at Hour 11.

This list is objectively defined by:

- public 1969 construction records,
 - Berlin archives,
 - and the 1997 maintenance documentation, reflected archival metadata:
“55 city names were added, 1 removed, spellings were partially changed.”
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4. Deterministic Key Table

Each entry yields one key character.

The full 27-letter block used for Row 3 is:

K L G K O R N A

Q Y I G H E M F

X V V G Z

U U C E Q N

(Spaces added for readability.)

This sequence comes directly from the Weltzeituhr error cities:

1–8: Russia block (Kiew → Alma Ata)

9–16: Global block (Quebec → Frunze)

17–21: Pacific block (Xian → Zagreb)

22–27: Return to Asia/Europe/Americas (Ulan Bator → Novosibirsk)

5. Decryption of Row 3

Ciphertext (31 chars)

V T T M Z F P K W G D K Z X T J C D I G K U H U A U E K C A R

5.1 Russia Block (Positions 1–8)

Key: K L G K O R N A

Plain: LINCLOCK

5.2 Global Block (Positions 9–16)

Key: Q Y I G H E M F

Plain: GIVESTHE

5.3 Pacific Block (Positions 17–21)

Key: X V V G Z

Plain: FINAL

5.4 Answer Block (Positions 22–27)

Key: U U C E Q N

Plain: ANSWER

5.5 Trailing Nulls

Final four ciphertext characters K C A R are treated as padding/nulls.

6. Decryption of Rows 1 and 2

OBKR's remaining letters direct simple Caesar shifts:

- **Row 1** uses **O** (15) → shift –15
→ “FACE THE DIAL TO THE EAST NORTH EAST”
- **Row 2** uses **B** (2) → shift –2
→ “LOOK TO THE SATURN RING AND REMEMBER”

These are validated by features of the Weltzeituhr:

- **Wind rose** (Row 1): ENE alignment matches the base plate.

- **Saturn** (Row 2): The clock's planetary indicator designates Saturn as the visually trackable cue.
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7. Full Plaintext

FACE THE DIAL TO THE EAST NORTH EAST.

LOOK TO THE SATURN RING AND REMEMBER:

BERLIN CLOCK GIVES THE FINAL ANSWER.

8. Reproducibility

Any cryptanalyst with:

- the 1969 and 1997 city lists,
- the Weltzeituhr geometry,
- the OBKR header, and
- the K4 ciphertext,

will derive the same key stream and the same plaintext.

End of Document.

Yours sincerely,

Ryan Bonifacino